

Seven-Strand T0 Crossing Matrices of Positive Pure Braids: An Exact Exhaustive Resolution

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Abstract

Let $C(b)_{ij}$ count positive crossings at which strand i passes over strand j in a positive braid diagram. We prove that every nonnegative, symmetric, zero-diagonal 7×7 integer matrix M satisfying

$$M_{ij} = M_{jk} = 0 \implies M_{ik} = 0 \quad (i < j < k)$$

is $C(b)$ for a positive pure seven-strand braid b . The finite part of the proof is an exhaustive certificate: all 2^{21} Boolean supports are tested, precisely 96,428 satisfy the condition, and an explicit positive pure Artin word is checked for each of them. The CN-support lifting theorem of Ozawa–Shimizu–Yaguchi, followed by an explicit positive-crossing assignment, then handles arbitrary multiplicities. Thus the unbounded integer statement is reduced to, and resolved by, an exact finite computation together with the published support reduction.

1 Statement and context

Let $B_7 = \langle \sigma_1, \dots, \sigma_6 \rangle$ be the seven-strand Artin braid group, let B_7^+ be its positive monoid, and let P_7 be the kernel of the endpoint-permutation map $B_7 \rightarrow S_7$. The strands are labelled $1, \dots, 7$ from left to right at the top. For a braid diagram b , let $n_{ij}^+(b)$ and $n_{ij}^-(b)$ denote the numbers of positive and negative crossings, respectively, at which strand i is over strand j . Its crossing matrix is

$$C(b)_{ij} = n_{ij}^+(b) - n_{ij}^-(b), \quad C(b)_{ii} = 0.$$

For a positive word, these entries are nonnegative directed crossing counts.

We say that a zero-diagonal matrix satisfies T0 if

$$(M_{ij} = 0 \text{ and } M_{jk} = 0) \implies M_{ik} = 0 \quad (1 \leq i < j < k \leq 7). \quad (\text{T0})$$

Theorem 1. *For every $M \in \text{Mat}_7(\mathbb{Z}_{\geq 0})$ that is symmetric, has zero diagonal, and satisfies T0, there is a braid $b \in B_7^+ \cap P_7$ such that $C(b) = M$.*

Gutierrez and Nitecki developed this crossing-matrix framework, studied positive realization, and gave a finite realization algorithm for a fixed matrix [1]. Ozawa, Shimizu, and Yaguchi recast the support problem through crossing-number (CN) matrices, proved the corresponding characterization through six strands, and recorded that there are 96,428 seven-strand Boolean T0 patterns [2]. The contribution here is to realize every one of those seven-strand patterns by an exactly checked word and then to lift the result directly to arbitrary integer entries.

Braid-word algebra also occurs in exact knot-invariant computations, for example in the structured two-bridge setting of [3]. That adjacent work concerns Links–Gould positivity, not reconstruction of a braid from a prescribed crossing matrix. A bounded literature search located the two crossing-matrix sources above but is not, by itself, a claim of priority or of previously established open status.

2 The Boolean certificate

For $1 \leq i < j \leq 7$, associate a bit to the unordered pair $\{i, j\}$, using lexicographic order

$$(1, 2), (1, 3), \dots, (1, 7), (2, 3), \dots, (6, 7).$$

Thus a symmetric Boolean zero-diagonal matrix is represented by a 21-bit mask. For each of the $\binom{7}{3} = 35$ triples $i < j < k$, the mask is rejected exactly when

$$S_{ij} = S_{jk} = 0, \quad S_{ik} = 1.$$

The accompanying certificate file `realizations.tsv` has one row consisting of a decimal mask and a word over the alphabet $\{1, \dots, 6\}$; the letter p denotes σ_p . The empty word is used for the zero mask.

Proposition 2. *Every symmetric Boolean zero-diagonal 7×7 matrix satisfying T0 is the crossing matrix of a positive pure seven-strand Artin word.*

Proof. The exact verifier iterates through the entire integer interval $0, \dots, 2^{21} - 1$. It tests all 35 implications above and requires the resulting ascending sequence of admissible masks to agree row-for-row with the certificate table. The scan finds 96,428 admissible masks and checks 96,428 distinct rows; therefore no admissible mask is omitted and no inadmissible mask is included.

For each row, the verifier starts with the label order $[1, 2, \dots, 7]$. On reading p , it records one directed crossing from the label currently in position p to that in position $p + 1$, and then swaps those two labels. If the mask has e set bits, the word is required to have length $2e$. At the end, the verifier requires the original label order, and hence purity. For every selected pair $\{i, j\}$ it requires directed counts

$$C_{ij} = C_{ji} = 1,$$

and for every unselected pair it requires both counts to vanish. All letters are positive Artin generators. These tests pass for all 96,428 rows, comprising 2,336,072 generator letters. Each row is consequently a direct positive certificate for its mask, and the complete mask scan proves the proposition. $\square$

This proof does not infer realizability from the search procedure that found the words. It uses only full mask coverage and direct simulation of the listed words. A separately written checker also performs a bitmap membership scan of all 2^{21} masks and the same labelled-strand simulation; it obtains zero missing or extra masks and zero word failures.

3 Lifting a support to arbitrary entries

For a braid projection B (a braid diagram with the over/under information forgotten), let $N(B)_{ij}$ be the total number of crossings between labelled strands i and j . Thus, if B underlies a positive pure word w , then

$$N(B) = C(w) + C(w)^T.$$

We use the following support-lifting consequence of Proposition 7 and the targetwise construction in the proof of Proposition 27 of Ozawa–Shimizu–Yaguchi [2].

Lemma 3 (CN-support lifting). *Let S be a symmetric Boolean zero-diagonal matrix. If a positive pure word w_S satisfies $C(w_S) = S$, then every symmetric nonnegative integer matrix M with*

$$S_{ij} = 1 \iff M_{ij} > 0$$

is the crossing matrix of a positive pure word.

Proof. Forget the over/under information in w_S . The resulting pure projection B_S has CN matrix

$$N(B_S) = S + S^T = 2S.$$

Represent symmetric CN matrices by their strictly upper triangular halves. The upper half of $2M$ is nonnegative and even, and the matrix obtained by replacing each of its nonzero entries by 2 is the upper half of $2S$. Proposition 7 of [2] therefore supplies a pure braid projection B_M with full symmetric CN matrix $2M$.

It remains only to restore crossing information. For fixed labelled strands i and j , let $B_M(i, j)$ count the crossings at which i moves from the left of j to its right. Every crossing reverses their relative order, and purity makes that order the same at the top and bottom. Hence their $2M_{ij}$ crossings split equally:

$$B_M(i, j) = B_M(j, i) = M_{ij}.$$

Choose the over/under information at every crossing so that the crossing is positive. With the sign convention used here, strand i moves left to right exactly when it passes under strand j . Consequently the number of crossings at which i is over j is $B_M(j, i) = M_{ij}$. This holds for every pair, so the resulting diagram is positive and pure and has crossing matrix M . This is precisely the targetwise transfer used in the proof of Proposition 27 of [2]. $\square$

Proof of Theorem 1. Let S be the Boolean support of M . Condition T0 depends only on which entries vanish, so S is a Boolean T0 matrix. Proposition 2 supplies a positive pure word w_S with $C(w_S) = S$. Lemma 3 expands this word to one with crossing matrix M . $\square$

Together with the known necessity of nonnegativity, symmetry, zero diagonal, and T0 for positive pure braids [1, 2], Theorem 1 gives the seven-strand characterization

$$C(B_7^+ \cap P_7) = \{M \in \text{Mat}_7(\mathbb{Z}_{\geq 0}) : M = M^T, M_{ii} = 0, \text{ and } M \text{ satisfies T0}\}.$$

4 Reproducibility and limitations

The certificate and verifier sources are supplied as ancillary files with the manuscript. The certificate is generated by the accompanying C++ source `sweep.cpp`. Its logical validation is performed by `verify_realizations.py`; a separately written replay program is the second implementation. From the project directory, the canonical generation and check commands are

```
g++ -std=c++20 -O3 -DNDEBUG -Wall -Wextra -pedantic \
  evidence/sweep.cpp -o evidence/sweep
evidence/sweep evidence/realizations.tsv
python3 evidence/verify_realizations.py \
  evidence/realizations.tsv evidence/verification.json
```

The exhaustive search took place only in the Boolean seven-strand layer. No sampled numerical evidence, floating-point calculation, symmetry quotient, or negative-search inference enters the proof. As with any computational proof, reproduction assumes correct execution of the displayed source on the stated input; Proposition 7 of [2] and the explicit positive-crossing transfer above handle all unbounded entries.

Computational and AI assistance. Computer programs generated the witness table and exhaustively checked mask coverage, endpoint permutations, and directed crossing counts. AI assistance was used in proof development, implementation and cross-checking of the exact verifiers, organization of the audit record, and manuscript preparation. The conclusion rests on the inspectable certificate, deterministic verifier source, and the explicit lifting proof, rather than on an unaudited AI output.

References

- [1] Mauricio Gutierrez and Zbigniew Nitecki. Crossing matrices of positive braids, 2018. arXiv:1805.12189 [math.GT]. URL: <https://arxiv.org/abs/1805.12189>, arXiv:1805.12189, doi:10.48550/arXiv.1805.12189.
- [2] Yuko Ozawa, Ayaka Shimizu, and Yoshiro Yaguchi. The CN matrix of a pure braid projection, 2025. arXiv:2506.08659v3 [math.GT]. URL: <https://arxiv.org/abs/2506.08659v3>, arXiv:2506.08659, doi:10.48550/arXiv.2506.08659.
- [3] Zijian Zeng. Links–Gould positivity for two-block alternating 2-bridge knots. SSRN preprint, 2026. SSRN 7385078. doi:10.2139/ssrn.7385078.